The square maximizes the first Steklov eigenvalue among parallelograms

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Abstract

We prove that among nondegenerate planar parallelograms with fixed perimeter, the square uniquely maximizes the first positive Steklov eigenvalue. The argument uses explicit trial functions and rigorous estimates for the constants in the square spectrum. The result addresses a restricted class in AIM 022; the polygonal optimization problem remains open.

Repository target. This manuscript concerns AIM problem 022 at revision `3fa50884d129`. Its scope is stated below.

Scope. The following proves the perimeter-normalized Steklov conjecture among all parallelograms, with strict rigidity. It does **not** settle the full catalogued conjecture for arbitrary convex quadrilaterals or for all polygons. A bounded literature search found the known rectangle theorem and work on inverse Steklov spectra of parallelograms, but I make no claim that this restricted result is new.

Theorem

Among all nondegenerate planar parallelograms of a given perimeter, the square uniquely maximizes the first positive Steklov eigenvalue.

Notation and elementary constants

Let $Q = (-1, 1)^2$, and let k be the unique root in $(0, \pi/2)$ of

$$\tan k \, \tanh k = 1.$$

Put $h = \tanh k$ and $s = kh$. The first positive Steklov eigenvalue of Q is s; its eigenspace is generated by

$$f(x, y) = \sin(kx)\cosh(ky), \qquad g(x, y) = \cosh(kx)\sin(ky).$$

The square's scale-invariant value is $8s$. The square spectral formula is standard; it also follows from the Robin separation argument below.

Introduce

$$\delta = \frac{k^2(1-h^4)+h^2}{2kh}, \qquad \beta = \frac{h}{k} - h^2, \qquad \kappa = \frac{1+h^4}{2kh}.$$

We need only the coarse inequalities

$$\delta > \frac{4}{5}, \qquad 0 < \beta < \frac{1}{3}, \qquad \kappa > \frac{2}{3}, \qquad 8s > \frac{224}{45} > 2\sqrt{3}. \tag{C}$$

These are elementary rigorous inequalities, not numerical assumptions. For example, Taylor series with their usual alternating or positive-tail bounds give

$$\frac{93}{100} < k < \frac{94}{100}, \qquad \frac{73}{100} < h < \frac{74}{100}.$$

Substituting the appropriate endpoints in the displayed formulas proves (C): use

$$\delta \geq \frac{(93/100)^2(1-(74/100)^4)+(73/100)^2}{2(94/100)(74/100)},$$

$$\beta \leq \frac{74}{93} - \left(\frac{73}{100}\right)^2, \qquad \kappa \geq \frac{1+(73/100)^4}{2(94/100)(74/100)}, \qquad 8s > 8\frac{93}{100}\frac{73}{100}.$$

The accompanying `parallelogram_constant_certificate.py` proves every bracket and comparison by exact rational arithmetic. For transparency, its exponential enclosure is the degree-20 Taylor sum plus a geometric bound on the positive tail, while sine and cosine use alternating sums through degree 17 and 16, respectively, followed by the next negative term. The root function $\sin x \tanh x - \cos x$ is strictly increasing on $(0, \pi/2)$, so its rationally enclosed signs at 93/100 and 94/100 certify the unique root bracket.

Parametrization

Scale the parallelogram so that its perimeter is eight. Translate its center to the origin. After interchanging and reversing edge vectors when necessary, it has the form

$$P = \{xv + yw : (x,y) \in Q\}, \qquad |v| = a = 1+t, \quad |w| = b = 1-t,$$

with $0 \leq t < 1$, and the angle between v, w belongs to $(0, \pi/2]$. Set

$$c = \cos\theta \in [0,1), \qquad z = \sin\theta = \sqrt{1-c^2}.$$

The area is $4abz$. Since the boundary is centrally symmetric, every linear coordinate function centered at the origin has zero boundary mean.

Two linear trial estimates

For any vector e, the boundary second moment is

$$\int_{\partial P} (e \cdot X)^2 \, ds = \frac{4}{3}(a+3b)(e\cdot v)^2 + \frac{4}{3}(3a+b)(e\cdot w)^2. \tag{1}$$

This follows by integrating separately along the four sides. Taking e to be a unit vector parallel to the longer edge and dropping the second nonnegative term gives

$$8\sigma_1(P) \le \frac{12(1-t)}{(1+t)(2-t)}. \tag{2}$$

Here the factor $z \le 1$ was discarded. The right side decreases with t; for $t \ge 1/8$, it is at most $224/45 < 8s$. This proves strict optimality outside $t < 1/8$.

Alternatively take e to be the unit angular bisector. Its scalar products with v, w are $a\cos(\theta/2), b\cos(\theta/2)$. Formula (1) then yields

$$8\sigma_1(P) \le 6(1-t^2)\sqrt{\frac{1-c}{1+c}} \le 6\sqrt{\frac{1-c}{1+c}}. \tag{3}$$

Thus for $c \ge 1/2$, the right side is at most $2\sqrt{3} < 8s$. It remains to treat

$$0 \le t \le \frac{1}{8}, \qquad 0 \le c \le \frac{1}{2}. \tag{4}$$

Exact two-dimensional trial space

Pull the square functions f, g to P by the affine map $(x, y) \mapsto xv + yw$. Both functions have boundary mean zero, and their boundary cross term vanishes by parity.

For clarity define

$$A_1 = k^2 \int_{-1}^{1} \cos^2(kx)dx \int_{-1}^{1} \cosh^2(ky)dy, \qquad A_2 = k^2 \int_{-1}^{1} \sin^2(kx)dx \int_{-1}^{1} \sinh^2(ky)dy,$$

$$d_a = 2\cosh^2 k \int_{-1}^{1} \sin^2(kx)dx, \qquad d_b = 2\sin^2 k \int_{-1}^{1} \cosh^2(ky)dy.$$

Let $E_0 = A_1 + A_2$ and $B_0 = d_a + d_b$. Direct integration, using $\tan k \tanh k = 1$, gives

$$E_0 = sB_0, \quad \frac{A_1 - A_2}{E_0} = \delta, \quad \frac{d_b - d_a}{B_0} = \beta, \quad \frac{\cosh(2k) - \cos(2k)}{E_0} = \kappa.$$

Under the affine transformation, the energy and boundary-norm matrices of the trial pair, normalized by E_0 and B_0, respectively, are

$$\mathcal{E} = \frac{1}{z}\begin{pmatrix} \dfrac{1+t^2-2\delta t}{1-t^2} & -\kappa c \\ -\kappa c & \dfrac{1+t^2+2\delta t}{1-t^2} \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} 1-\beta t & 0 \\ 0 & 1+\beta t \end{pmatrix}. \tag{5}$$

The mixed energy uses

$$\int_Q (f_x g_y + f_y g_x)\, dxdy = \cosh(2k) - \cos(2k),$$

whereas the other mixed integrals vanish by parity. Formula (5) is exact.

Case A: $c \geq 8t^2$

Use the trial function $f + g$. Formula (5) and (C) give

$$\frac{\sigma_1(P)}{s} \leq \frac{(1+t^2)/(1-t^2) - \kappa c}{z}. \tag{6}$$

For $t \leq 1/8$,

$$\frac{1+t^2}{1-t^2} \leq 1 + \frac{128}{63}t^2.$$

For $c \leq 1/2$,

$$1 - z \leq (2 - \sqrt{3})c.$$

Moreover

$$\frac{2}{3} - (2 - \sqrt{3}) = \sqrt{3} - \frac{4}{3} > \frac{1}{3}.$$

Therefore, if $c \geq 8t^2$ and $(c, t) \neq (0, 0)$,

$$\kappa c - (1 - z) > \frac{c}{3} \geq \frac{8}{3}t^2 > \frac{128}{63}t^2$$

when $t > 0$; for $t = 0, c > 0$ the same strict comparison follows directly. Hence the numerator in (6) is strictly smaller than z, and $\sigma_1(P) < s$.

Case B: $c < 8t^2$

This forces $t > 0$. Use f alone. Formula (5) gives

$$\frac{\sigma_1(P)}{s} \leq \frac{1 + t^2 - 2\delta t}{z(1-t^2)(1-\beta t)}. \tag{7}$$

By (C), the factor excluding $1/z$ is bounded above by

$$\frac{1 + t^2 - \frac{8}{5}t}{(1-t^2)(1-t/3)} < 1 - t. \tag{8}$$

To verify the strict inequality in (8), subtract its numerator from $(1-t)(1-t^2)(1-t/3)$. The result is

$$t\left(\frac{4}{15} - \frac{5}{3}t + \frac{4}{3}t^2 - \frac{1}{3}t^3\right) > 0$$

for $0 < t \leq 1/8$, because the bracket is at least $4/15 - 5/24 - 1/1536 > 0$.

Finally,

$$z = \sqrt{1 - c^2} \geq 1 - c^2 > 1 - 64t^4 > 1 - t$$

on this range. Inserting this into (7)–(8) proves $\sigma_1(P) < s$.

The only remaining case is $t = c = 0$, which is the square and gives equality. Scaling back proves the theorem.

Why the square formula is the first eigenvalue

This paragraph can replace a citation if a self-contained account is preferred. Consider the one-dimensional Robin operator on $(-1, 1)$ with outward boundary condition $\partial_\nu u - su = 0$. Since $0 < s < 1$, it has exactly one negative eigenvalue, $-k^2$, with eigenfunction $\cosh(kx)$. Its first positive eigenvalue is k^2, with eigenfunction $\sin(kx)$, because $k \tanh k = k \cot k = s$. The remaining eigenvalues exceed k^2 by the one-dimensional Sturm oscillation theorem.

On Q, tensor products therefore give exactly one negative eigenvalue of the Robin Laplacian, namely $-2k^2$, and zero is its next eigenvalue, of multiplicity two. Its zero eigenfunctions are f, g above. Equivalently, the quadratic form $q(u) = \int_Q |\nabla u|^2 - s \int_{\partial Q} u^2$ has negative index one. If a mean-zero-boundary test function had $q(u) < 0$, then $q(1) < 0$ and $q(1, u) = -s \int_{\partial Q} u = 0$, producing a two-dimensional negative subspace, a contradiction. Thus all such functions have Rayleigh quotient at least s, while f, g attain s.

Sources and limitations

The benchmark square/rectangle formulas and general cuboid separation method are treated in A. Girouard, J. Lagacé, I. Polterovich and A. Savo, *The Steklov spectrum of cuboids*, Mathematika **65** (2019), 272–310, doi:10.1112/S0025579318000414; arXiv:1711.03075v2. The present affine trial-space proof is given in full above, so its validity does not depend on an unquoted theorem about general parallelograms.

The argument uses central symmetry twice: to ensure the trial functions' boundary means vanish and to obtain the two-parameter affine representation. A general quadrilateral has extra geometric parameters and is not an affine square. Consequently this theorem is a restricted result for catalogued problem 022, not a full resolution of that entry.